



Munahil Amin

Works on the collective's computer-vision and classification projects.



2 projects · part of AI Collective

PROJECTS

CO-AUTHOR

 Source

Computer vision · PyTorch · Jan 2026

GAN-balanced plant disease classification ↗

Does generating minority-class images fix an imbalanced dataset?

A GAN generates extra leaf images for the under-represented classes of a 38-class plant disease dataset. A custom CNN is then trained on the balanced set and checked with 5-fold cross-validation.

ACCURACY

98.19%

MACRO F1

0.982

CLASSES

38

GAN

CNN

Class imbalance

K-fold

[Dataset on Kaggle ↗](#)

TEAM MEMBER

 Source

Particle physics data · scikit-learn · Nov 2025

HIGGS: logistic regression vs neural network [↗](#)

How much does model capacity matter at 11 million samples?

Both models were trained on 16 slices of the HIGGS collision dataset (11M events, 28 features), from 1% to 100%, to find where extra data stops paying off. The network's accuracy is already 76.6% at 30% of the data and only reaches 76.8% at 100%.

ROC-AUC

0.852 LR 0.684

ACCURACY

76.8% LR 64.1%

TRAIN TIME

3,204s LR 87s

MLP

Logistic regression

Scaling study

ROC / PR

[Report \(PDF\)](#) [↗](#)